How much of breast cancer grade is in the routine record? A leakage-audited, multi-cohort open-data prediction study with MRI radiomics and transport to Saudi case-mix

Running title: Breast cancer grade from routine variables: open multi-cohort study

Author
Dr. Aminat Abiola Ajibola[1,*]

[1]Assistant Professor, College of Computer Science and Engineering, University of Hafr Al-Batin, Eastern Province, Kingdom of Saudi Arabia.

[*]**Corresponding author:** Dr. Aminat Abiola Ajibola, College of Computer Science and Engineering, University of Hafr Al-Batin, Eastern Province, Kingdom of Saudi Arabia. Tel.: +966-550047554. E-mail: aajibola@uhb.edu.sa

CRediT authorship contribution statement: Aminat Abiola Ajibola: Conceptualization, Methodology, Software, Formal analysis, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. [Add co-authors if any.]

Funding: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. [Confirm.]

Declaration of competing interest: The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. [Confirm.]

Declaration of generative AI and AI-assisted technologies in the writing process: During the preparation of this work the author used Claude (Anthropic) to assist with analysis code, reference checking and drafting of the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. [Confirm or edit.]

Acknowledgements: [Optional.]

Ethics: Secondary analysis of de-identified, publicly released data; no approval or consent required.